Lab Manual: Notebook 5 — Finding Patterns in Brain and Behavioral Data

1. Introduction: The Workflow Design Framework

1.1. The Ask → Build → Document Cycle

This manual implements the **Ask → Build → Document** cycle, the professional standard for integrating AI assistants into scientific research. In this framework, you are the **Workflow Designer**. Your responsibility is to provide strategic direction to AI tools, verify the scientific and statistical integrity of the generated code, and meticulously document your software environment to ensure your findings are reproducible by others.

1.2. ⚠️ Crucial Scientific Guardrail

Preventing Overfitting When applying machine learning to brain data, the most significant risk is "Overfitting." This occurs when a model is tested on the same data it used for training, leading to results that appear perfect but fail to predict anything on new data.**Designer Note:** AI assistants often default to the simplest possible code, which may accidentally omit a strict data split. You must explicitly command the AI to maintain a "blind" testing set and verify that no training data has leaked into your evaluation phase.

1.3. Project Context

- **Initiative Title:** Scientific Workflows for Brain and Behavioral Research
- **Funding:** National Science Foundation Supported Learning Initiative (Award No. OAC-2417875)

2. Key Terminology

Term,Definition
Spike Sorting,The process of separating mixed electrical recordings from multiple neurons into individual sources based on the unique shape of each neuron's electrical discharge (spike).
Neural Decoding,"The use of statistical learning to identify patterns in brain activity that predict specific behaviors, intentions, or cognitive states."
Software Libraries (Dependencies),"Pre-built collections of code (e.g., NumPy, scikit-learn) that provide standardized mathematical functions, allowing researchers to build complex workflows without writing every algorithm from scratch."

3. Lab Setup and Profile Sync

3.1. Student Identification

Initialize your session by completing the Python block below. **Note:** Ensure you edit only the text within the quotation marks to avoid syntax errors.

```
student_name = "Your Name"
institution = "Your University/Organization"
active_notebook = "Notebook 05 — Pattern Recognition and Analysis"
```

3.2. System Verification

Execute the profile sync cell in your Google Colab/Jupyter environment. This records your progress and confirms that your computational environment is correctly configured.

4. Part A: Identifying Neurons and Behavioral Intentions

4.1. Topic Overview

In this section, you will master two pillars of computational neuroscience: **Signal Separation** (untangling the activity of individual cells) and **Behavioral Decoding** (predicting intent from those signals). These processes are the foundation of modern Brain-Machine Interfaces (BMIs).

4.2. Activity A.1 — Neural Signal Separation

Direct your AI assistant to generate a signal analysis script using the prompt below.**AI Prompt:**"Act as a computational neuroscience instructor explaining neural signal analysis to a biology student. Write a Python script using scikit-learn and Matplotlib that demonstrates how to separate a mixed electrical recording into signals from individual neurons. The script should:

1. Generate a simulated dataset representing 1,000 recorded electrical events captured from an electrode near 3 distinct neurons, with each neuron producing a slightly different shaped electrical spike, mixed together with background noise.
2. Use Principal Component Analysis (PCA) to find the most distinctive features of each spike shape, reducing the data to its two most informative dimensions.
3. Apply a K-means clustering algorithm to group the spikes into 3 clusters.
4. Display a scatter plot of the clusters and the average spike shape for each neuron. Include plain-language comments explaining the utility of this separation for understanding brain activity."✅ **Validation Check:** Before proceeding, verify that your generated plot shows exactly three distinct color clusters and three corresponding mean spike-shape traces.

4.3. Activity A.2 — Predicting Behavioral Intentions

Use the following prompt to build a predictive model.**AI Prompt:**"Act as a machine learning instructor teaching a neuroscience student how to decode behavioral intentions from brain signals. Write a Python script using scikit-learn that:

1. Creates a simulated dataset of 500 trials, each with 10 signal features (e.g., frequency band power) labeled by intended action (e.g., 'Move Left' vs. 'Move Right').
2. Splits the data into an 80% training set and a 20% testing set.
3. Trains a classifier (SVM or Random Forest) using ONLY the training data.
4. Evaluates the model on the testing data and outputs an accuracy score and a confusion matrix. Include comments explaining why keeping training and testing data separate is critical for an honest evaluation."✅ **Validation Check:** Ensure the script outputs a numerical accuracy score. Identify the confusion matrix—can you see where the model made its most frequent errors?

5. Part B: Reproducible and Secure Workflows

5.1. Topic Overview

The final phase of the workflow cycle is **Documentation** . To be scientifically valid, your analysis must be reproducible. This requires tracking the exact software versions used and managing sensitive configuration settings securely to prevent unauthorized access to research data.

5.2. Activity B.1 — Software Environment Documentation

Use this prompt to generate a reproducibility and security script.**AI Prompt:**"Act as a research reproducibility instructor. Write a Python script that:

1. Checks for a configuration setting stored as an environment variable. If missing, print a warning and use a safe default.
2. Prints the exact version numbers of NumPy, SciPy, and scikit-learn.
3. Demonstrates secure handling of a sensitive access key by retrieving it from a hidden system variable, and illustrates the security risk by showing what would happen if a developer accidentally printed the secret value directly. Use comments to explain why these practices are essential for collaborative research."✅ **Validation Check:** In the script's output, locate the version number for **scikit-learn** . Write this number down; you will need it for your reflection.

6. Integration and Reflection

6.1. Part A Reflection

- **Visual Distinction:** In Activity A.1, how effectively did PCA and K-means separate the three neurons? Were there overlapping points that suggest "noise" in the signal?
- **Predictive Power:** What was the classifier's accuracy on the unseen data?
- **Critical Concept Connection:**
- **Overfitting (Concepts 22 & 23):** Explain why testing a decoder on the same data used for training results in a "misleadingly optimistic" evaluation.
- **BMI Utility (Concepts 24 & 25):** How does the ability to decode intent (Activity A.2) and separate individual neurons (Activity A.1) enable the use of **Sensory Feedback** and **Neural Plasticity** in prosthetic design?

6.2. Part B Reflection

- **Configuration Logic:** Describe the script's behavior when the environment variable was missing. How does this prevent the workflow from crashing?
- **Security & Reproducibility:** Referencing **Concepts 44, 46, and 48** , why is it a violation of research security to hard-code passwords? How does documenting library versions (as you did in Activity B.1) solve the problem of a collaborator getting different results on their machine?

7. Reference Checklist and Self-Check

7.1. Concept Alignment Table

Record the Cell ID (e.g., "Activity A.1 Code Cell") where you successfully demonstrated each concept.| Concept Area | Concept Number | Where You Demonstrated It (Cell ID) || ------ | ------ | ------ || Neuroscience | Concept 22 — Spike sorting | || Neuroscience | Concept 23 — Machine learning for neural decoding | || Neuroscience | Concept 24 — Sensory feedback | || Neuroscience | Concept 25 — Neural plasticity | || Computing | Concept 44 — Hidden configuration settings | || Computing | Concept 46 — Code version tracking (Git) | || Computing | Concept 48 — Data array | |

7.2. Specific Concept Review

- **Concept 22: Spike Sorting:** Separating mixed electrical recordings from multiple neurons into individual signals based on distinct discharge shapes.
- **Concept 23: Machine Learning for Neural Decoding:** Using algorithms to find patterns in brain signals that predict behavior or intent.
- **Concept 24: Sensory Feedback:** Delivering information (e.g., electrical stimulation) back to the user to simulate touch in a prosthetic system.
- **Concept 25: Neural Plasticity:** The brain's ability to adapt and reorganize, allowing users to improve device control over time.
- **Concept 44: Hidden Configuration Settings:** Storing sensitive values (like API keys) in the operating system rather than the code.
- **Concept 46: Code Version Tracking (Git):** A system for recording changes to code to allow collaboration and recovery of previous versions.
- **Concept 48: Data Array:** A structured list of numbers that serves as the fundamental building block for efficient scientific calculation.

7.3. Next Steps

You have completed the foundational pattern recognition training. You are now prepared for **Notebook 6: The Capstone** , where you will integrate all four stages of the research workflow into a single, automated end-to-end pipeline.